GGG-A-821C
September 23, 1987
SUPERSEDING
GGG-A-821B
July 27, 1960

FEDERAL SPECIFICATION

AUGER, PIPE AND SEWER, HAND; AND AUGER, WATER-CLOSET, HAND

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers flexible coil spring wire, closet, and pipe and sewer augers, and flat spring steel, pipe and sewer augers used for opening clogged pipes and closet bowls.

1.2 Classification.

1.2.1 Types, classes and styles. Closet and pipe and sewer augers covered by this specification shall be of the following types, classes and styles, as specified (see 6.1):

Type I. Auger, water-closet
Type IV. Auger, pipe and sewer

Class 1. Flexible coil spring (offset sliding handle)
Class 2. Top snake
Style A - Light duty
Style B - Heavy duty

Class 3. Flat spring steel

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-W-428 - Wire, Steel, High Carbon, Round, Uncoated for Mechanical Springs, (General Purpose)
PPP-P-40 - Packaging and Packing Of Hand Tools

Federal Standards:

Federal Standard No. 123 - Marking for Domestic Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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(Single copies of this specification, and other Federal specifications and commercial item desciption required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation-Packaging, Methods Of

MIL-S-46049 - Steel, Carbon Strip, Cold Rolled, Hardened and Tempered Spring Quality

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards

ASTM A228 - Steel Wire, Music Spring Quality

ASTM D3951 - Standard Practice for Commercial Packaging

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein.

3.1.1 Coil springs. The coil springs shall be made of wire conforming to Type I or II of Federal Specification QQ-W-428 or music wire conforming to ASTM A228.

3.1.2 Flat springs. The flat springs shall be made of flat spring strip conforming to material number 1, regular temper of MIL-S-46049.

3.2 Finish.

3.2.1 Coil springs. Coil springs shall be smoothly finished and in accordance with the finish requirements of Federal specification QQ-W-428 or ASTM A228.

3.2.2 Flat springs. Flat springs shall be smoothly finished and in accordance with the finish requirements of MIL-S-46049.

3.2.3 Component parts. Component parts made of ferrous metal, other than the coil spring, shall be smoothly finished and shall be free from sharp edges and treated to resist corrosion.

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3.3 Identification marking. Each auger shall be etched or stamped in a permanent and legible manner with the manufacturer's name or identifying symbol of such known character that the source may be readily identifiable, part number and country of origin.

3.4 Type I auger, water-closet. Type I auger shall consist of a wire coil spring, hook-type springhead, metal tube, rubber or plastic bumper, fastening device, crankshaft, with a handle to revolve on the crankshaft and shall be furnished with coil spring lengths of 3, 6, or 9 feet, as specified (see 6.1).
Type I auger shall be similar to figure 1.

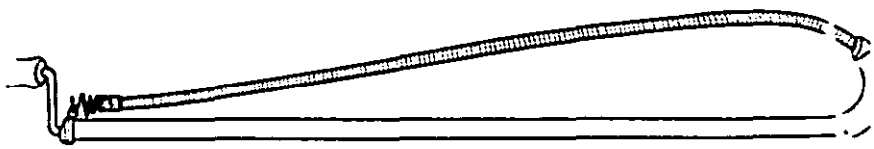


Figure 1. Type I, Auger, Water-closet

3.4.1 Coil spring. Coil springs made of music wire shall be 0.100 inch minimum diameter, wound to 7/16 ± 0.025 inch outside diameter coil. Coil springs made of carbon steel wire shall be 0.120 inch minimum diameter, wound to 1/2 ± 0.025 inch outside diameter coil.

3.4.2 Hook head. The hook-type head shall be designed so that it may be separated easily and replaced with the slightest effort and it shall bore through as well as entangle and remove the obstruction without becoming loose during the operation. The head may be either of the stationary or drop head type, at the option of the manufacturer.

3.4.3 Tube. The tube shall be of brass, steel, or aluminum. Brass tubes shall be heavy-gage, seamless and tempered to prevent crushing, marring or scratching. Steel tubes shall be either plated or enameled coated. Aluminum tubes shall be treated to resist corrosion. The tubes shall have a return bend to catch and hold water from spilling when removing the spring from the bowl and shall be made of brass, steel, or aluminum.

3.4.4 Fastening device. There shall be a fastening device securely fastened to the handle end of the tube for holding the hook head when not in use.

3.4.5 Crankshaft. The crankshaft shall be a one piece cold rolled steel bar or rod extending into the wire coil and fastened in a manner to prevent mechanical failure.

3.4.6 Handle. There shall be a handle attached to the crankshaft affording a adequate handgrip and revolve freely on the crankshaft.

3.5 Type IV auger, pipe and sewer.

3.5.1 Class 1, flexible coil spring (offset sliding handle). Class 1 auger shall consist of a flexible coil spring wire, cleaning head and sliding reversible handle.

3.5.1.1 Flexible coil spring wire. The flexible coil spring wire shall be either carbon steel or music wire and either right-hand or left-hand coiled.

3.5.1.2 Cleaning head. The cleaning head shall be of the bulbous type so designed that it may be replaced, without loss of the entire spring, by cutting the spring and screwing a new head on the remaining spring. The cleaning head shall be reinforced so that it will not come loose during a cleaning operation.

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3.5.1.3 Handle. The handle shall be of the reversible offset type, made of iron or steel, with either one or two loose brass or corrosion resistant steel grips that will provide a full grip for the operator's hand. The handle shall be capable of sliding along the coil spring and gripping it tightly at any point by means of a clamping method in such a manner that a pull of 100 pounds on the handle will not cause either slippage along the coil spring, mechanical failure of the clamping means or handle, or noticeable deformation of the spring.

3.5.1.4 Type IV, class 1 augers shall be similar to figure 2 and shall conform to table I for the diameter and length specified (see 6.1).

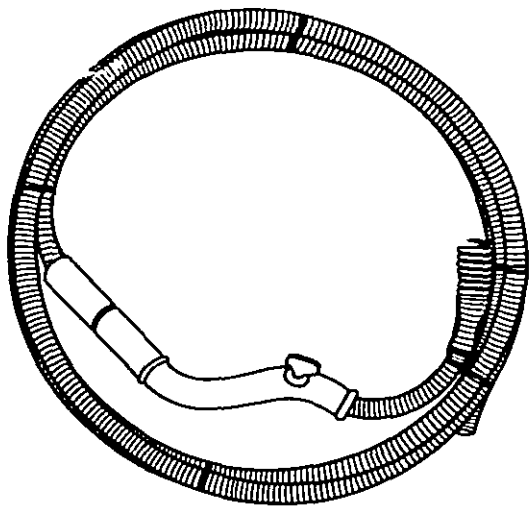


Figure 2. Type IV, class 1, pipe and sewer auger, flexible coil spring (offset sliding handle)

TABLE I. Type IV, class 1, pipe and sewer auger, flexible coil spring (offset sliding handle).

Nominal Outside Diameter (Inch)	Outside Diameter of Coil Springs: Actual Outside Diameter (Inch) — Music Wire Construction Min.	Music Wire Construction Max.	Carbon Steel Wire Construction Min.	Carbon Steel Wire Construction Max.	Wire Diameter (Inch) — Music Wire	Carbon Steel Wire Min.	Carbon Steel Wire Max.	Length Of Coil Springs (Min) (Feet)
1/4	0.240	0.260	0.225	0.305	0.076	0.086	0.091	10, 15 or 25
3/8	0.365	0.385	0.350	0.400	0.090	0.113	0.120	10, 15, 25, 35 or 50
7/16	0.4275	0.4475	-	-	0.121	-	-	15, 25, 35, 50, 75 or 100

3.5.2 Class 2, top snake. Class 2 augers shall consist of an enclosed housing, a handcrank, a locking device, and a coil spring with a cleaning head attached.

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3.5.2.1 Enclosed housing. The housing body shall be made of aluminum, steel, or a high impact grade plastic. Aluminum housing bodies shall be not less than 0.040 inch thick; steel not less than 0.030 inch thick; and plastic housing bodies not less than 0.200 inch thick for style A auger, or not less than 0.080 inch thick for style B augers. A cover shall be securely fastened to the housing body.

3.5.2.2 Handcrank. The auger shall have a handcrank for revolving the housing and spring. The handcrank shall be fitted with a revolving handgrip.

3.5.2.3 Locking device. All class 2 augers shall be fitted with a device for positive locking of the coil spring to the housing without permanent set or damage to the coil spring.

3.5.2.4 Coil spring. The coil spring may be either right-hand or left-hand wound. Unless otherwise specified (see 6.1), one length of coil, with cleaning head attached, shall be furnished with each class 2 auger. The cleaning head may be either of the stationary or drop head type, at the option of the manufacturer.

3.5.2.5 Style A, light duty. Style A augers shall have the handcrank securely fastened to the cover and shall not extend beyond the outside diameter of the housing. Style A augers shall be similar to figure 3 and shall conform to table II for the diameter and length of coil spring specified (see 6.1).

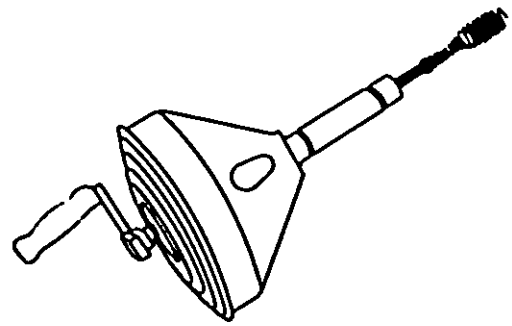


Figure 3. Type IV, class 2, style A, pipe and sewer auger, top snake, light duty.

TABLE II. Type IV, class 2, style A, pipe and sewer auger, top snake, light duty.

Nominal Outside Diameter (Inch)	Outside Diameter of Coil Springs: Actual Outside Diameter (Inch) — Music Wire Construction (Min)	(Max)	Carbon Steel Wire Construction (Min)	(Max)	Wire Diameter (Inch) — Music Wire	Carbon Steel Wire (Min)	(Max)	Length Of Coil Springs (Min) (Feet)	
1/4	0.225	0.275	0.225	0.305	0.076	0.086	0.091	15 or 25	
3/8	0.350	0.400	0.350	0.400	0.090	0.113	0.120	25 or 35	

3.5.2.6 Style B. heavy duty. Style B augers shall have a base, which may be set on the floor, in or upon which the housing will turn freely while being operated in an upright position. The auger shall be revolved either by crank handle attached to the back of the container or by an independently hand-operated, fully enclosed, geared, revolver type device attached to the coil spring. Style B augers shall be similar to figure 4 and shall conform to table III for the diameter and length of coil spring specified (see 6.1).

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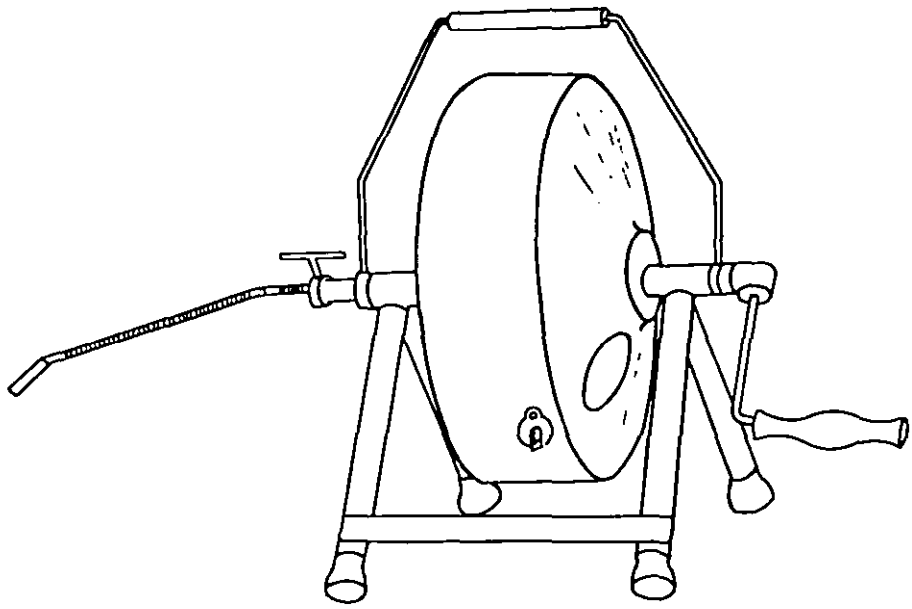


Figure 4. Type IV, class 2, style B, pipe and sewer auger, top snake, heavy duty.

TABLE III. Type IV, class 2, style B, pipe and sewer auger, top snake, heavy duty.

Outside Diameter of Coil Springs								
Nominal Outside Diameter (Inch)	Actual Outside Diameter (Inch)				Wire Diameter (Inch)			Length Of Coil Springs (Min) (Feet)
	Music Wire Construction		Carbon Steel Wire Construction			Carbon Steel Wire		
	(Min)	(Max)	(Min)	(Max)	Music Wire	(Min)	(Max)	
1/2	-	-	0.475	0.540	-	0.135	0.148	50, 75 or 100

3.5.3 Class 3, flat spring steel. Class 3 augers shall consist of a flat spring steel auger of 50 feet minimum length, one inch wide, 0.125 inch thick, one sliding metal handle, one spear-point head assembled on one end, one revolving roller head assembled on the opposite end and a heavy steel frame constructed in such a manner that the auger may be coiled and stored ready for use. Class 3 augers shall be similar to figure 5.

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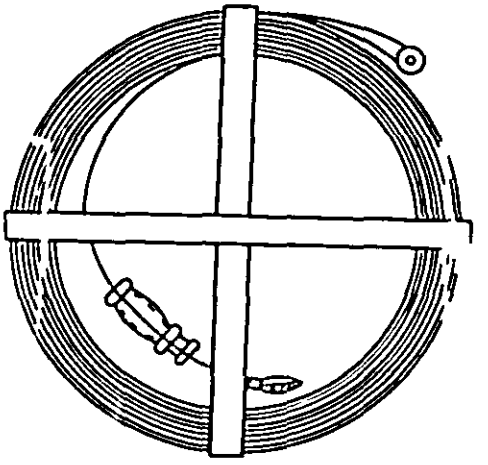


Figure 5. Type IV, class 3, pipe and sewer auger, flat spring steel.

3.6 Workmanship. Augers shall be uniform and free from all defects affecting the appearance or which might adversely affect their durability and serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be in accordance with MIL-STD-105, except where otherwise indicated. The lot size shall consist of all augers of the same type, style, and size manufactured by the same process and submitted for inspection at the same time. The sample unit is one auger.

4.2.1 Sampling for examination for visual and dimensional defects. Each auger shall be examined in accordance with paragraph 4.3 and for dimensional defects. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

4.2.2 Sampling for testing. Each auger shall be tested in accordance with paragraph 4.4. The inspection level shall be S-2 with an AQL of 2.5 percent defective.

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4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined in accordance with the defects set forth in the applicable subparagraphs at the inspection level and acceptable quality levels (AQL's) specified. Random samples shall be drawn from each lot of the items for examination of visual, dimensional, and preparation for delivery defects.

4.3.2 Examination for visual and dimensional defects. Any auger in the sample lot containing one or more visual or dimensional defects shall be rejected, and if the number of defective augers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. The inspection level shall be S-2 with an AQL of 4.0 percent defective. The auger shall be examined for the defects shown in Table IV.

TABLE IV. Classification of defects in accordance with MIL-STD-105.

Categories	Defects
Critical:	
1	None defined.
Major:	
101.	Type, class, style, and size not as specified.
102.	Incomplete, component parts missing.
103.	Component parts materials not as specified.
104.	Coil spring broken, wire not smoothly finished.
105.	Cleaning head defective, not easily removed and replaced or not prevented from becoming loose during service.
106.	Handle not free of sharp edges or burrs, not a comfortable grip, not free to revolve, or handle not prevented from becoming detached from crank-shaft.
107.	Tube fastening device missing or not secure, tube not of sufficient proportions to prevent crushing, not provided with a return bend, or tube not smooth and round with square-cut ends (type I only).
108.	Crankshaft not one piece, or not suitably fastened to the coil spring (type I only).
109.	Handle fails to slide on spring, clamp fails to secure handle anywhere along spring, or clamp deforms spring (type IV, class 1 only).
110.	Enclosed housing cover not securely fastened to the body, handcrank extends beyond outside diameter of housing, locking device fails to securely hold spring, or locking device distorts or puts a permanent set in the coil spring (type IV, class 2, styles A and B only).
111.	Base defective, or housing fails to turn on base (type IV, class 2, style B only).
112.	Coil springs diameter and length not as specified.
113.	Component parts not smoothly finished, evidence of sharp edges, or parts not treated to preclude corrosion.
114.	Auger not easily rotated; auger sticks or binds, inoperable.
Minor:	
201.	None defined.

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4.4 Testing. Each sample unit shall be tested in accordance with paragraph 4.4. The inspection level shall be S-2 with AQL of 4.0 percent defective.

4.4.1 Coil springs. From each sample taken in accordance with 4.2.3, having the wire size, a test specimen long enough to provide material for this test but not more than 3 feet long shall be removed for chemical analysis, tension test and wrapping test in accordance with Federal Specification QQ-W-428 or ASTM A228.

4.4.2 Flat spring steel. From each sample taken in accordance with 4.2.3 a test specimen long enough to provide material for this test shall be removed for the bending test in accordance with MIL-S-46049.

4.4.3 Possible test failures. Possible test failures are defined as follows:

(a) Coil springs:
Failure of coil spring sample to meet chemical analysis, tension, and wrapping requirements.
(b) Flat springs:
Failure of flat spring sample to meet bending requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or Commercial, as specified (see 6.1).

5.1.1 Level A. Level A preservation and packaging shall be in accordance with Level A of PPP-P-40.

5.1.2 Commercial. The augers shall be packaged to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling and storage. (DOD may use Commercial Packaging Standard ASTM D 3951).

5.2 Packing. Packing shall be level A, or commercial as specified (see 6.1).

5.2.1 Level A. Level A packing shall be in accordance with level A of PPP-P-40.

5.2.2 Commercial packing. Augers preserved and packaged in accordance with 5.1.1 or 5.1.2 as specified (see 6.2) shall be packed in a manner to insure delivery at destination, to provide for redistribution by the initial receiving activity, and shall be acceptable to common carrier under the Uniform Freight Classification and the National Motor Freight Classification Rules. (DOD may use Commercial Packaging Standard ASTM D3951).

5.3 Marking. Marking shall be as specified in the contract or order.

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6. NOTES

6.1 *Ordering data.* Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class and style required (see 1.2).
(c) Size required (see 3.4, 3.5.1.4, 3.5.2.5 and 3.5.2.6, tables I, II and III).
(d) If more than one length of coil is required (see 3.5.2.4).
(e) Diameter and length of coil spring required (see 3.5.2.6).
(f) Applicable levels of preservation, packaging, and packing required (see 5.1 and 5.2).

MILITARY INTERESTS:

Military Coordinating Activity

Army - GL

Custodians

Army - GL
Air Force 99

Review Activity

Air Force - 84

PREPARING ACTIVITY:

GSA - FSS

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